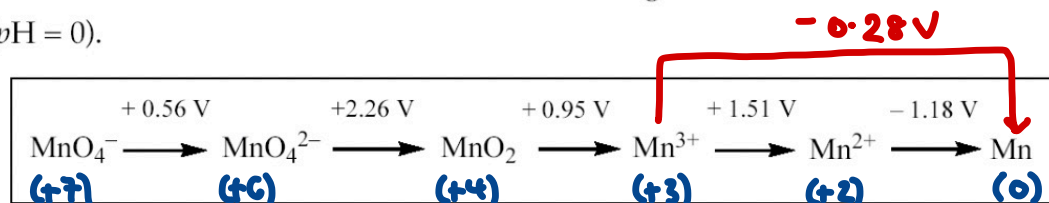
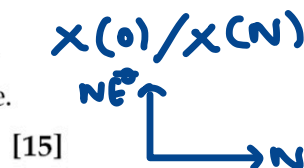


Use the following Latimer diagram, which shows the standard reduction potentials $E^\ominus / \text{V}$ for some oxidation states of manganese in acidic solution ($\text{pH} = 0$).



- (i) Construct a Frost (oxidation state) diagram for manganese at $\text{pH} = 0$ and identify species that you would predict from this diagram to be unstable.



- (ii) Calculate the equilibrium constant for the disproportionation reaction of Mn^{3+} at $\text{pH} = 0$.

[10]

[Note that at $T = 298 \text{ K}$, $RT/F = 0.026 \text{ V}$ or $(RT/F)\ln 10 = 0.059 \text{ V}$]



$$\therefore 0.85 F = -3 F \times E^\ominus \quad \therefore \underline{\underline{E^\ominus = -0.28 \text{ V}}} \quad (\text{Mn}^{3+}/\text{Mn}^0)$$

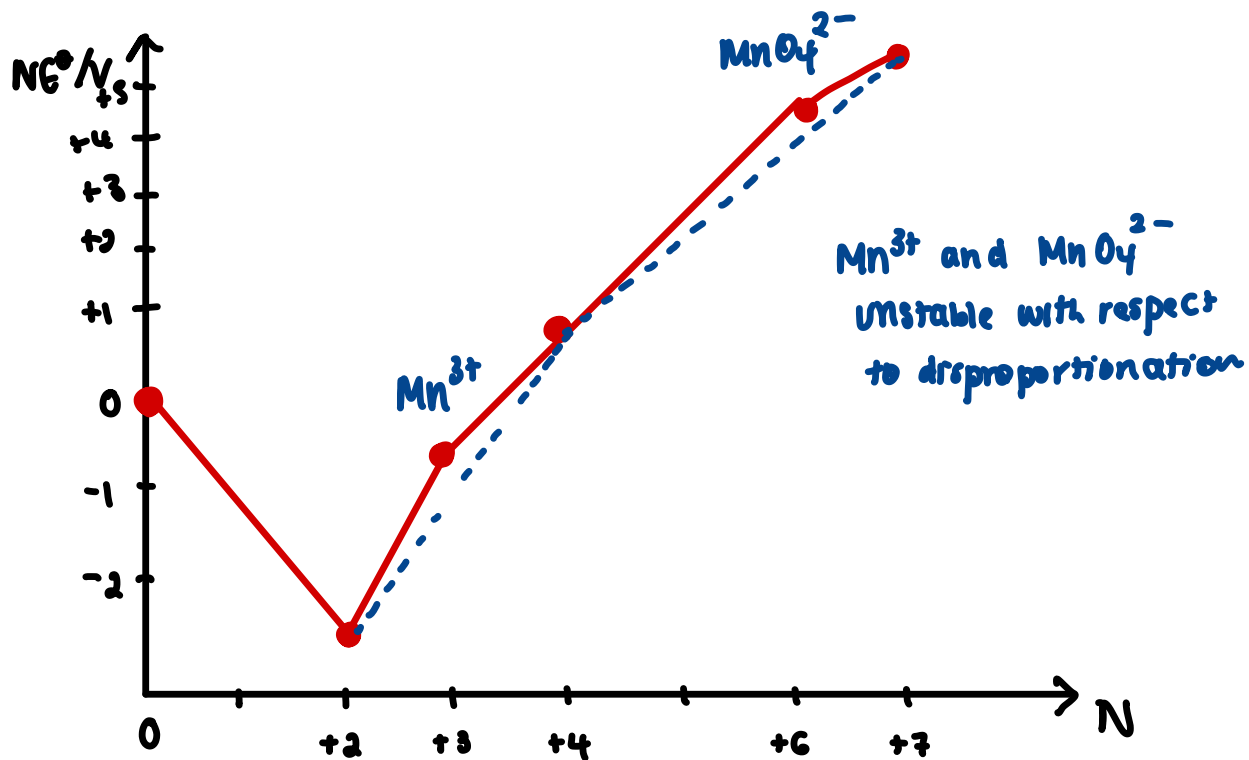


$$-0.1 F = -4 F \times E^\ominus \quad \therefore \underline{\underline{E^\ominus = 0.025 \text{ V}}} \quad (\text{Mn}^{4+}/\text{Mn}^0)$$

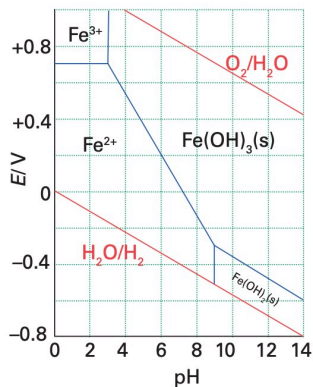
$$E^\ominus (\text{MnO}_4^{2-}/\text{Mn}) = 0.77 \text{ V}$$

$$E^\ominus (\text{MnO}_4^-/\text{Mn}) = 0.74 \text{ V}$$

N	0	+2	+3	+4	+6	+7
NE°	0	-2.36	-0.85	+0.1	+4.62	+5.18



Pourbaix diagrams



Shows conditions of potential and pH under which species are stable in water

- Vertical lines → species related by proton transfer only
- Horizontal lines → species related by electron transfer only
- Sloped lines → species related by both H^+/e^- transfer

Nernst equation

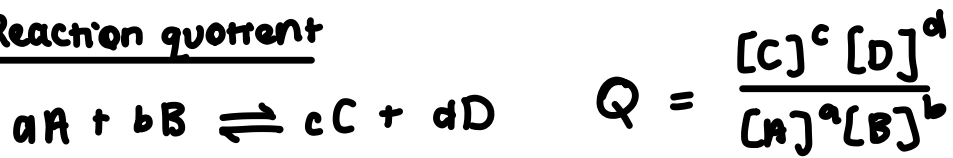
$$E = E^\circ - \frac{RT}{nF} \ln Q$$

standard potential

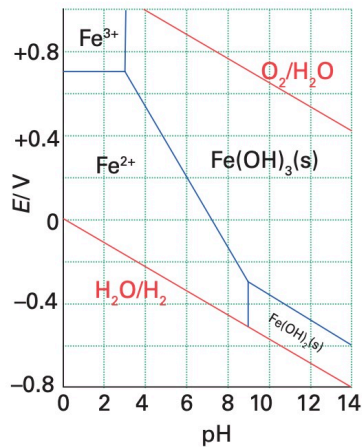
non-standard potential

Reaction quotient

Reaction quotient



Measure of relative amounts of reactants/products in a reaction at any given time (not necessarily equilibrium)



What is the slope of the Pourbaix

diagram for $\text{Fe(OH)}_3 + 3\text{H}^+ + \text{e}^- \rightarrow \text{Fe}^{2+}$?

$$E = E^\circ - \frac{RT}{F} \ln \left(\frac{[\text{Fe}^{2+}]}{[\text{Fe(OH)}_3] [\text{H}^+]^3} \right)$$

assume $[\text{Fe}^{2+}] = [\text{Fe(OH)}_3]$

$$\begin{aligned} E &= E^\circ - \frac{RT}{F} \ln \left(\frac{1}{[\text{H}^+]^3} \right) = E^\circ + \frac{3RT}{F} \ln(\text{H}^+) \\ &= E^\circ + \frac{3RT}{F} (-\text{pH} \times \ln 10) = E^\circ - \underbrace{\frac{3RT}{F} \ln 10}_{=-3 \times 0.0595} \times \text{pH} \end{aligned}$$

Changing bases

$$\log_b N = y \rightarrow \text{convert to base } a$$

$$N = b^y$$

$$\boxed{\log_a N} = \log_a b^y = y \log_a b = \boxed{\log_b N \times \log_a b}$$

Change $\ln$ to $\log_{10}$

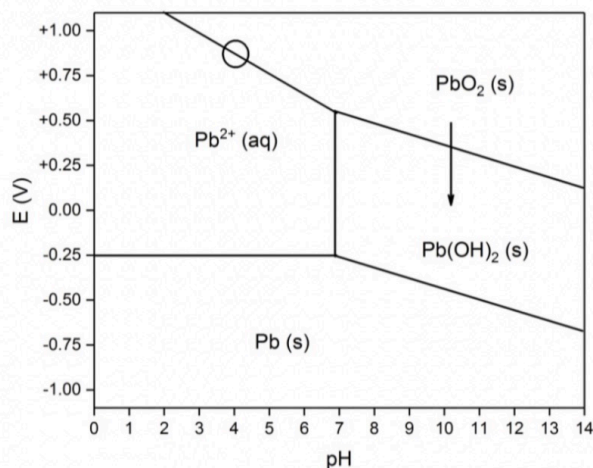
$$\log_e N = y \quad N = e^y$$

$$\log_{10} N = y \log_{10} e = \log_e N \times \boxed{\log_{10} e}$$

change $\log_{10}$ to $\log_e$

$$\log_{10} N = y \quad N = 10^y$$

$$\log_e N = y \log_e 10 = \log_{10} N \times \boxed{\log_e 10}$$



$$E = E^\circ - \frac{RT}{2F} \ln \left(\frac{1}{[\text{H}^+]^4} \right)$$

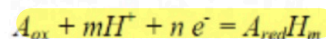
$$= E^\circ + \frac{4RT}{2F} \ln(\text{H}^+)$$

$$= E^\circ + \frac{2RT}{F} (-\text{pH} \times \ln 10)$$

$$= E^\circ - \frac{2RT}{F} \ln 10 \times \text{pH}$$

(iii) Derive an expression for the slope marked with a circle in the diagram from the Nernst equation. [4]

$$\text{slope} = -2 \times 0.0595 \\ = \underline{\underline{-0.118}}$$

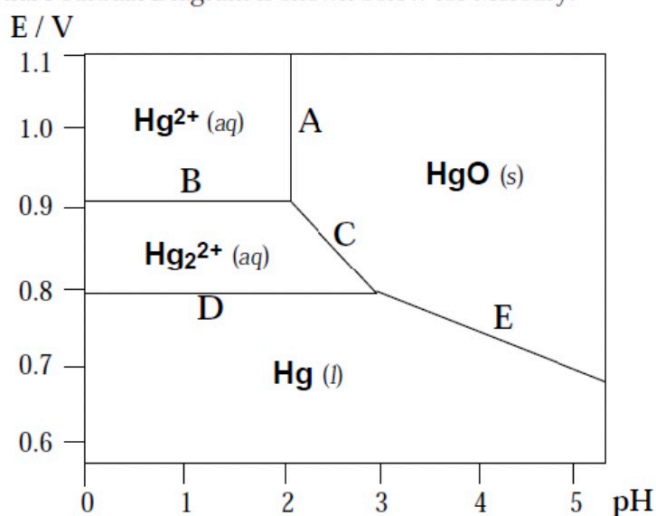


$$E = E^\circ + 0.059/n \log \{ [A_{\text{ox}}] / [A_{\text{red}}\text{H}_m] \} - 0.059 m/n \times \text{pH}$$

$$E_{\text{pH}} = E^\circ - 0.059 \text{ pH } (m/n)$$

Slope = $m/n \times -0.059 \text{ pH}$; $m=4$ and $n=2$; slope = $-0.118...$ match with the measured slope from the figure.

(b) A partial Pourbaix Diagram is shown below for Mercury.



(i) Write balanced equations for each of the reactions or half-reactions corresponding to lines A to E in the diagram. [5]

(ii) Explain the different slopes of lines C and E, and predict the expected ratio of these slopes. [4]